

Practical 6

Problem definition:

You are a Data Analyst at ShopSmart Analytics Pvt. Ltd. The company operates a large-scale e-commerce platform that processes thousands of transactions every day. Management wants to identify high-performing product categories, top-selling products, and revenue contribution by city. Traditional spreadsheet tools are unable to efficiently process the growing volume of transaction data. Therefore, the company has adopted Apache Spark for scalable analytics. Your task is to design a Spark-based analytical solution to process sales data and generate business insights.

Objectives:

1. To load e-commerce sales data into a Spark DataFrame.
2. To display and validate the schema and data types of the dataset.
3. To create a Revenue column using Quantity $\times$ Price.
4. To calculate category-wise revenue.
5. To identify top-selling products based on quantity sold.
6. To calculate city-wise revenue.
7. To sort categories based on revenue generated.
8. To display analytical summaries.
9. To interpret business insights and provide suitable recommendations.

Observation / Output:

```

tushar@Tushar-laptop: ~/Desktop/practical6_BDA
(tushar@Tushar-laptop) ~/Desktop/practical6_BDA
/home/tushar/Desktop/practical6_BDA/spark-3.5.7-bin-hadoop3
bash: /home/tushar/Desktop/practical6_BDA/spark-3.5.7-bin-hadoop3: Is a directory

(tushar@Tushar-laptop) ~/Desktop/practical6_BDA
$ ./spark-3.5.7-bin-hadoop3/bin/spark-submit --version
26/09/24 22:11:46 WARN Utils: Your hostname, tushar-laptop resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)
26/09/24 22:11:46 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Welcome to
Spark version 3.5.7

Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 21.0.11
Branch HEAD
Compiled by user runner on 2025-09-17T20:37:30Z
Revision ed00b46951a7ecd6a6429accd3b9c5b2dc792b65
URL https://github.com/apache/spark
Type --help for more information.

(tushar@Tushar-laptop) ~/Desktop/practical6_BDA
$ export SPARK_HOME=$HOME/Desktop/practical6_BDA/spark-3.5.7-bin-hadoop3

(tushar@Tushar-laptop) ~/Desktop/practical6_BDA
$ export PATH=$SPARK_HOME/bin:$PATH

(tushar@Tushar-laptop) ~/Desktop/practical6_BDA
$ spark-submit --version
26/09/24 22:12:16 WARN Utils: Your hostname, Tushar-laptop resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (on interface lo) because ye error hai aur final practical
26/09/24 22:12:16 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Welcome to
Spark version 3.5.7

Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 21.0.11
Branch HEAD
Compiled by user runner on 2025-09-17T20:37:30Z
Revision ed00b46951a7ecd6a6429accd3b9c5b2dc792b65
URL https://github.com/apache/spark
Type --help for more information.
  
```

Task 1 — Load Sales Data into Spark DataFrame

from pyspark.sql import SparkSession

```

spark = SparkSession.builder \
    .appName("E-Commerce Sales Analysis") \
    .master("local[*]") \
    .getOrCreate()
  
```

```
df = spark.read.csv(
```

```
"sales_data.csv",
header=True,
inferSchema=True
)
```

```
df.show(10)
```

```
spark.stop()
```

```
(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
ls -lh
total 383M
drwxr-xr-x 13 tushar tushar 4.0K Sep 18 2025 spark-3.5.7-bin-hadoop3
-rw-r--r-- 1 tushar tushar 383M Sep 18 2025 spark-3.5.7-bin-hadoop3.tgz

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
ls -lh
total 383M
-rw-r--r-- 1 tushar tushar 103K Sep 24 22:21 sales_data.csv
drwxr-xr-x 13 tushar tushar 4.0K Sep 18 2025 spark-3.5.7-bin-hadoop3
-rw-r--r-- 1 tushar tushar 383M Sep 18 2025 spark-3.5.7-bin-hadoop3.tgz

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
head -5 sales_data.csv
Product_ID,Sale_Date,Sales_Rep,Region,Sales_Amount,Quantity,Sold,Product_Category,Unit_Cost,Unit_Price,Customer_Type,Discount,Payment_Method,Sales_Channel,Region_and_Sales_Rep
1052,2023-02-03,Bob,North,5053.97,18,Furniture,152.75,267.22,Returning,0.09,Cash,Online,North-Bob
1053,2023-04-21,Bob,West,4384.02,17,Furniture,3816.39,4209.44,Returning,0.11,Cash,Retail,West-Bob
1015,2023-09-21,David,South,4531.23,38,Food,241.56,321.4,Returning,0.2,Same_Transfer,Retail,South-David
1072,2023-08-24,Bob,South,2167.94,39,Clothing,4330.03,4467.75,New,0.02,Credit Card,Retail,South-Bob

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
nano sales_analysis.py

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
spark-submit sales_analysis.py
26/09/24 22:25:37 WARN Utils: Your hostname, Tushar-Laptop resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)
26/09/24 22:25:37 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/09/24 22:25:38 INFO SparkContext: Running Spark version 3.5.7
26/09/24 22:25:38 INFO SparkContext: OS Info Linux, 6.18.33.2-microsoft-standard-WSL2, amd64
26/09/24 22:25:38 INFO SparkContext: Java version 21.0.11
26/09/24 22:25:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/09/24 22:25:39 INFO ResourceUtils:
26/09/24 22:25:39 INFO ResourceUtils: No custom resources configured for spark.driver.
26/09/24 22:25:39 INFO ResourceUtils:
26/09/24 22:25:39 INFO SparkContext: Submitted application: E-Commerce Sales Analysis
26/09/24 22:25:39 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, am
ount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpu, amount: 1.0)
26/09/24 22:25:39 INFO ResourceProfile: Limiting resource is cpu
26/09/24 22:25:39 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/09/24 22:25:39 INFO SecurityManager: Changing view acls to: tushar
26/09/24 22:25:39 INFO SecurityManager: Changing modify acls to: tushar
26/09/24 22:25:39 INFO SecurityManager: Changing view acls groups to:
26/09/24 22:25:39 INFO SecurityManager: Changing modify acls groups to:
26/09/24 22:25:39 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: tushar; groups with modify permissions: tushar; groups with modify perm
issions: EMPTY
26/09/24 22:25:39 INFO Utils: Successfully started service 'sparkDriver' on port 34803.
26/09/24 22:25:39 INFO SparkEnv: Registering MapOutputTracker
26/09/24 22:25:39 INFO SparkEnv: Registering BlockManagerMaster
26/09/24 22:25:39 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/09/24 22:25:39 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/09/24 22:25:39 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/09/24 22:25:39 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-694be644-fcfb-4302-83ff-a89d9d5d1a8
26/09/24 22:25:39 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/09/24 22:25:39 INFO SparkEnv: Registering OutputCommitCoordinator
26/09/24 22:25:39 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/09/24 22:25:39 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/09/24 22:25:39 INFO Executor: Starting executor ID driver on host 10.255.255.254
```

```
tushar@Tushar-Laptop: ~/Desktop/practical6_BDA
Session Actions Edit View Help

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
nano sales_analysis.py

(tushar@ Tushar-Laptop):~/Desktop/practical6_BDA
spark-submit sales_analysis.py
26/09/24 22:28:39 WARN Utils: Your hostname, Tushar-Laptop resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)
26/09/24 22:28:39 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/09/24 22:28:40 INFO SparkContext: Running Spark version 3.5.7
26/09/24 22:28:40 INFO SparkContext: OS Info Linux, 6.18.33.2-microsoft-standard-WSL2, amd64
26/09/24 22:28:40 INFO SparkContext: Java version 21.0.11
26/09/24 22:28:40 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/09/24 22:28:40 INFO ResourceUtils:
26/09/24 22:28:40 INFO ResourceUtils: No custom resources configured for spark.driver.
26/09/24 22:28:40 INFO ResourceUtils:
26/09/24 22:28:40 INFO SparkContext: Submitted application: E-Commerce Sales Analysis
26/09/24 22:28:40 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, am
ount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpu, amount: 1.0)
26/09/24 22:28:40 INFO ResourceProfile: Limiting resource is cpu
26/09/24 22:28:40 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/09/24 22:28:40 INFO SecurityManager: Changing view acls to: tushar
26/09/24 22:28:40 INFO SecurityManager: Changing view acls groups to:
26/09/24 22:28:40 INFO SecurityManager: Changing modify acls groups to:
26/09/24 22:28:40 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: tushar; groups with view permissions: EMPTY; users with modify permissions: tushar; groups with modify perm
issions: EMPTY
26/09/24 22:28:40 INFO Utils: Successfully started service 'sparkDriver' on port 34191.
26/09/24 22:28:40 INFO SparkEnv: Registering MapOutputTracker
26/09/24 22:28:40 INFO SparkEnv: Registering BlockManagerMaster
26/09/24 22:28:40 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/09/24 22:28:40 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/09/24 22:28:40 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/09/24 22:28:41 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-2003606b-8194-4d19-a7e6-9c40d47bd55e
26/09/24 22:28:41 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/09/24 22:28:41 INFO SparkEnv: Registering OutputCommitCoordinator
26/09/24 22:28:41 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/09/24 22:28:41 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/09/24 22:28:41 INFO Executor: Starting executor ID driver on host 10.255.255.254
26/09/24 22:28:41 INFO Executor: OS Info Linux, 6.18.33.2-microsoft-standard-WSL2, amd64
26/09/24 22:28:41 INFO Executor: Java version 21.0.11
26/09/24 22:28:41 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/09/24 22:28:41 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@675f9bdf for default.
26/09/24 22:28:41 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 45277.
26/09/24 22:28:41 INFO NettyBlockTransferService: Server created on 10.255.255.254:45277
26/09/24 22:28:41 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/09/24 22:28:41 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 10.255.255.254, 45277, None)
26/09/24 22:28:41 INFO BlockManagerMaster: Initialized BlockManager: BlockManagerId(driver, 10.255.255.254, 45277, None)
26/09/24 22:28:41 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/09/24 22:28:41 INFO SharedState: Warehouse path is 'file:/home/tushar/Desktop/practical6_BDA/spark-warehouse'.
26/09/24 22:28:42 INFO InMemoryFileIndex: It took 31 ms to list leaf files for 1 paths.
26/09/24 22:28:42 INFO InMemoryFileIndex: It took 2 ms to list leaf files for 1 paths.
26/09/24 22:28:44 INFO FileSourceStrategy: Post-Scan Filters: (length(trim(value0), None)) > 0
26/09/24 22:28:44 INFO FileSourceStrategy: Pushed Filters:
26/09/24 22:28:44 INFO CodeGenerator: Code generated in 145.165468 ms
26/09/24 22:28:44 INFO MemoryStore: Block broadcast_0_piece0 stored as values in memory (estimated size 399.9 KiB, free 434.2 MiB)
26/09/24 22:28:44 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 34.3 KiB, free 434.2 MiB)
26/09/24 22:28:44 INFO BlockManagerInfo: Added broadcast_0_piece0 in memory on 10.255.255.254:45277 (size: 34.3 KiB, free: 434.4 MiB)
```

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Session Actions Edit View Help
26/09/24 22:31:06 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on 10.255.255.254:33577 (size: 34.3 KiB, free: 434.3 MiB)
26/09/24 22:31:06 INFO SparkContext: Created broadcast 6 from showString at NativeMethodAccessorImpl.java:0
26/09/24 22:31:06 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194384 bytes, open cost is considered as scanning 4194384 bytes.
26/09/24 22:31:06 INFO SparkContext: Starting Job: showString at NativeMethodAccessorImpl.java:0
26/09/24 22:31:06 INFO DAGScheduler: Got job 3 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:31:06 INFO DAGScheduler: Final stage: ResultStage 3 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:31:06 INFO DAGScheduler: Parents of final stage: List()
26/09/24 22:31:06 INFO DAGScheduler: Missing parents: List()
26/09/24 22:31:06 INFO DAGScheduler: Submitting ResultStage 3 (MapPartitionsRDD[17] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:31:06 INFO MemoryStore: Block broadcast_7_piece0 stored as values in memory (estimated size 37.4 KiB, free 433.4 MiB)
26/09/24 22:31:06 INFO BlockManagerInfo: Added broadcast_7_piece0 in memory on 10.255.255.254:33577 (size: 8.2 KiB, free: 434.2 MiB)
26/09/24 22:31:06 INFO SparkContext: Created broadcast 7 from broadcast at DAGScheduler.scala:1011
26/09/24 22:31:06 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 3 (MapPartitionsRDD[17] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/09/24 22:31:06 INFO TaskSchedulerImpl: Adding task set 0.0 with 1 tasks resource profile 0
26/09/24 22:31:06 INFO TaskSetManager: Starting task 0.0 in stage 3.0 (TID 3) (10.255.255.254, executor driver, partition 0, PROCESS_LOCAL, 9619 bytes)
26/09/24 22:31:06 INFO Executor: Running task 0.0 in stage 3.0 (TID 3)
26/09/24 22:31:06 INFO CodeGenerator: Code generated in 10.470503 ms
26/09/24 22:31:06 INFO FileScanRDD: Reading file path: file:///home/tushar/Desktop/practical6_BDA/sales_data.csv, range: 0-105083, partition values: [empty row]
26/09/24 22:31:06 INFO CodeGenerator: Code generated in 5.697193 ms
26/09/24 22:31:06 INFO Executor: Finished task 0.0 in stage 3.0 (TID 3). 2116 bytes result sent to driver
26/09/24 22:31:06 INFO TaskSetManager: Finished task 0.0 in stage 3.0 (TID 3) in 40 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:31:06 INFO TaskSchedulerImpl: Removed TaskSet 3.0, whose tasks have all completed, from pool
26/09/24 22:31:06 INFO DAGScheduler: ResultStage 3 (showString at NativeMethodAccessorImpl.java:0) finished in 0.047 s
26/09/24 22:31:06 INFO DAGScheduler: Job 3 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:31:06 INFO TaskSchedulerImpl: Killing all running tasks in stage 3: Stage finished
26/09/24 22:31:06 INFO DAGScheduler: Job 3 finished: showString at NativeMethodAccessorImpl.java:0, took 0.049806 s
26/09/24 22:31:06 INFO CodeGenerator: Code generated in 11.435873 ms

+-----+-----+-----+-----+-----+
|Product_ID|Quantity_Sold|Unit_Price|Sales_Amount|Calculated_Revenue|
+-----+-----+-----+-----+-----+
|10521|181|267.221|5853.971|4889.9608000000001|
|10931|171|4209.441|4384.021|71568.481|
|10151|301|171.41|4631.231|11142.0|
|10721|91|4467.751|2187.961|176242.2151|
|10611|131|692.711|3758.21|9085.231|
|10211|31|1166.511|3761.11|35480.321|
|10831|291|2624.091|618.311|76898.611|
|10871|461|3964.651|7698.921|182373.91|
|10511|201895.44999999999999|422.391|32863.49999999999999|
|10751|181|2682.341|8239.581|48282.121|
+-----+-----+-----+-----+-----+

only showing top 10 rows

```

```

Session Actions Edit View Help
26/09/24 22:33:08 INFO DAGScheduler: looking for newly runnable stages
26/09/24 22:33:08 INFO DAGScheduler: running: Set()
26/09/24 22:33:08 INFO DAGScheduler: waiting: Set()
26/09/24 22:33:08 INFO DAGScheduler: failed: Set()
26/09/24 22:33:08 INFO ShufflePartitionsUtil: For shuffle(0), advisory target size: 67108864, actual target size 1048576, minimum partition size: 1048576
26/09/24 22:33:08 INFO CodeGenerator: Code generated in 11.472483 ms
26/09/24 22:33:08 INFO SparkContext: Starting Job: showString at NativeMethodAccessorImpl.java:0
26/09/24 22:33:08 INFO DAGScheduler: Got job 5 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:33:08 INFO DAGScheduler: Final stage: ResultStage 6 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:33:08 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 5)
26/09/24 22:33:08 INFO DAGScheduler: Missing parents: List()
26/09/24 22:33:08 INFO DAGScheduler: Submitting ResultStage 6 (MapPartitionsRDD[25] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:33:08 INFO MemoryStore: Block broadcast_10_piece0 stored as values in memory (estimated size 47.5 KiB, free 433.8 MiB)
26/09/24 22:33:08 INFO BlockManagerInfo: Added broadcast_10_piece0 in memory on 10.255.255.254:33535 (size: 22.1 KiB, free: 434.3 MiB)
26/09/24 22:33:08 INFO SparkContext: Created broadcast 10 from broadcast at DAGScheduler.scala:1011
26/09/24 22:33:08 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 6 (MapPartitionsRDD[25] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/09/24 22:33:08 INFO TaskSchedulerImpl: Adding task set 6.0 in stage 6.0 (TID 5) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:33:08 INFO TaskSetManager: Starting task 6.0 in stage 6.0 (TID 5) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:33:08 INFO Executor: Running task 6.0 in stage 6.0 (TID 5)
26/09/24 22:33:08 INFO CodeGenerator: Code generated in 8.926899 ms
26/09/24 22:33:08 INFO ShuffleBlockFetcherIterator: Getting 1 (536.0 B) non-empty blocks including 1 (336.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/24 22:33:08 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 9 ms
26/09/24 22:33:08 INFO CodeGenerator: Code generated in 11.997373 ms
26/09/24 22:33:08 INFO Executor: Finished task 6.0 in stage 6.0 (TID 5). 5281 bytes result sent to driver
26/09/24 22:33:09 INFO TaskSetManager: Finished task 6.0 in stage 6.0 (TID 5) in 110 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:33:09 INFO TaskSchedulerImpl: Removed TaskSet 6.0, whose tasks have all completed, from pool
26/09/24 22:33:09 INFO DAGScheduler: ResultStage 6 (showString at NativeMethodAccessorImpl.java:0) finished in 0.125 s
26/09/24 22:33:09 INFO DAGScheduler: Job 5 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:33:09 INFO TaskSchedulerImpl: Killing all running tasks in stage 6: Stage finished
26/09/24 22:33:09 INFO DAGScheduler: Job 5 finished: showString at NativeMethodAccessorImpl.java:0, took 0.137993 s
26/09/24 22:33:09 INFO CodeGenerator: Code generated in 8.526387 ms
26/09/24 22:33:09 INFO CodeGenerator: Code generated in 3.782999 ms

+-----+-----+
|Product_Category|Total_Revenue|
+-----+-----+
|Clothing|1.9286733929999999E7|
|Furniture|1.8338888888888888E7|
|Electronics|1.7571848570000000E7|
|Food|1.5141885649999999E7|
+-----+-----+

only showing top 10 rows

```

```

26/09/24 22:34:53 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twollevel.enabled is set to true, but current version of codegen fast hashmap does not support this aggregate.
26/09/24 22:34:53 INFO CodeGenerator: Code generated in 20.456818 ms
26/09/24 22:34:53 INFO SparkContext: Starting Job: showString at NativeMethodAccessorImpl.java:0
26/09/24 22:34:53 INFO DAGScheduler: Got job 7 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:34:53 INFO DAGScheduler: Final stage: ResultStage 9 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:34:53 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 8)
26/09/24 22:34:53 INFO DAGScheduler: Missing parents: List()
26/09/24 22:34:53 INFO DAGScheduler: Submitting ResultStage 9 (MapPartitionsRDD[33] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:34:53 INFO MemoryStore: Block broadcast_13_piece0 stored as values in memory (estimated size 46.1 KiB, free 433.9 MiB)
26/09/24 22:34:53 INFO BlockManagerInfo: Added broadcast_13_piece0 in memory on 10.255.255.254:46215 (size: 21.6 KiB, free: 434.2 MiB)
26/09/24 22:34:53 INFO SparkContext: Created broadcast 13 from broadcast at DAGScheduler.scala:1011
26/09/24 22:34:53 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 9 (MapPartitionsRDD[33] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/09/24 22:34:53 INFO TaskSchedulerImpl: Adding task set 9.0 with 1 tasks resource profile 0
26/09/24 22:34:53 INFO TaskSetManager: Starting task 9.0 in stage 9.0 (TID 7) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:34:53 INFO Executor: Running task 9.0 in stage 9.0 (TID 7)
26/09/24 22:34:53 INFO ShuffleBlockFetcherIterator: Getting 1 (5.3 KiB) non-empty blocks including 1 (5.3 KiB) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/24 22:34:53 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 2 ms
26/09/24 22:34:53 INFO CodeGenerator: Code generated in 19.19386 ms
26/09/24 22:34:53 INFO Executor: Finished task 9.0 in stage 9.0 (TID 7). 5520 bytes result sent to driver
26/09/24 22:34:53 INFO TaskSetManager: Finished task 9.0 in stage 9.0 (TID 7) in 74 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:34:53 INFO TaskSchedulerImpl: Removed TaskSet 9.0, whose tasks have all completed, from pool
26/09/24 22:34:53 INFO DAGScheduler: ResultStage 9 (showString at NativeMethodAccessorImpl.java:0) finished in 0.088 s
26/09/24 22:34:53 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:34:53 INFO TaskSchedulerImpl: Killing all running tasks in stage 9: Stage finished
26/09/24 22:34:53 INFO DAGScheduler: Job 7 finished: showString at NativeMethodAccessorImpl.java:0, took 0.100721 s
26/09/24 22:34:53 INFO CodeGenerator: Code generated in 14.577463 ms

+-----+-----+
|Product_ID|Total_Quantity_Sold|
+-----+-----+
|10901|590|
|10921|548|
|10621|508|
|10991|502|
|10881|452|
|10671|407|
|10311|404|
|10211|398|
|10611|391|
|10891|390|
+-----+-----+

only showing top 10 rows

```

```

Session Actions Edit View Help
26/09/24 22:36:33 INFO SparkContext: Starting Job: showString at NativeMethodAccessorImpl.java:0
26/09/24 22:36:33 INFO DAGScheduler: Got Job 9 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:36:33 INFO DAGScheduler: Final stage: ResultStage 12 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:36:33 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 11)
26/09/24 22:36:33 INFO DAGScheduler: Missing parents: List()
26/09/24 22:36:33 INFO DAGScheduler: Submitting ResultStage 12 (MapPartitionsRDD[41] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:36:33 INFO MemoryStore: Block broadcast_16_piece0 stored as bytes in memory (estimated size 22.1 KiB, free 433.8 MiB)
26/09/24 22:36:33 INFO BlockManagerInfo: Added broadcast_16_piece0 in memory on 10.255.255.254:46805 (size: 22.1 KiB, free: 434.3 MiB)
26/09/24 22:36:33 INFO SparkContext: Created broadcast 16 from broadcast at DAGScheduler.scala:1611
26/09/24 22:36:33 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 12 (MapPartitionsRDD[41] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partition
26/09/24 22:36:33 INFO TaskSchedulerImpl: Adding task set 12.0 with 1 tasks resource profile 0
26/09/24 22:36:33 INFO TaskSetManager: Starting task 0.0 in stage 12.0 (TID 9) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:36:33 INFO Executor: Running task 0.0 in stage 12.0 (TID 9)
26/09/24 22:36:33 INFO ShuffleBlockFetcherIterator: Getting 1 (320.0 B) non-empty blocks including 1 (320.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) r
26/09/24 22:36:33 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:36:33 INFO Executor: Finished task 0.0 in stage 12.0 (TID 9), 5265 bytes result sent to driver
26/09/24 22:36:33 INFO TaskSetManager: Finished task 0.0 in stage 12.0 (TID 9) in 16 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:36:33 INFO TaskSchedulerImpl: Removed TaskSet 12.0, whose tasks have all completed, from pool
26/09/24 22:36:33 INFO DAGScheduler: ResultStage 12 (showString at NativeMethodAccessorImpl.java:0) finished in 0.027 s
26/09/24 22:36:33 INFO DAGScheduler: Job 9 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:36:33 INFO TaskSchedulerImpl: Killing all running tasks in stage 12: Stage finished
26/09/24 22:36:33 INFO DAGScheduler: Job 9 finished: showString at NativeMethodAccessorImpl.java:0, took 0.033744 s

+-----+
|Region|      Total_Revenue|
+-----+
|North|1.8208403850000013E7|
|East|1.807740105000001E7|
|West|1.776156864999999E7|
|South|1.6282567160000017E7|
+-----+

26/09/24 22:36:33 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:36:34 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:36:34 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:36:34 INFO MemoryStore: MemoryStore cleared
26/09/24 22:36:34 INFO BlockManager: BlockManager stopped
26/09/24 22:36:34 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:36:34 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:36:34 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:36:34 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:36:34 INFO ShutdownHookManager: Deleting directory /tmp/spark-607bf260-fe76-4a81-9022-74594bdd94c9
26/09/24 22:36:34 INFO ShutdownHookManager: Deleting directory /tmp/spark-a955428c-34ac-4fc5-b421-205e7baf5e4d/pyspark-43ead37c-a65e-4f80-9b3d-56442f4f464a
26/09/24 22:36:34 INFO ShutdownHookManager: Deleting directory /tmp/spark-a955428c-34ac-4fc5-b421-205e7baf5e4d

(tushar@Tushar-Laptop): ~/Desktop/practical6_BDA

```

```

Session Actions Edit View Help
26/09/24 22:42:08 INFO DAGScheduler: failed: Set()
26/09/24 22:42:08 INFO CodeGenerator: Code generated in 5.403282 ms
26/09/24 22:42:08 INFO SparkContext: Starting Job: collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78
26/09/24 22:42:08 INFO DAGScheduler: Got Job 17 (collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78) with 1 output partitions
26/09/24 22:42:08 INFO DAGScheduler: Final stage: ResultStage 24 (collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78)
26/09/24 22:42:08 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 23)
26/09/24 22:42:08 INFO DAGScheduler: Missing parents: List()
26/09/24 22:42:08 INFO DAGScheduler: Submitting ResultStage 24 (MapPartitionsRDD[69] at collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78), which has no missing parents
26/09/24 22:42:08 INFO MemoryStore: Block broadcast_28 stored as values in memory (estimated size 14.0 KiB, free 433.6 MiB)
26/09/24 22:42:08 INFO MemoryStore: Block broadcast_28_piece0 stored as bytes in memory (estimated size 6.5 KiB, free 433.6 MiB)
26/09/24 22:42:08 INFO BlockManagerInfo: Added broadcast_28_piece0 in memory on 10.255.255.254:43623 (size: 6.5 KiB, free: 434.3 MiB)
26/09/24 22:42:08 INFO SparkContext: Created broadcast 28 from broadcast at DAGScheduler.scala:1611
26/09/24 22:42:08 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 24 (MapPartitionsRDD[69] at collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78) (first 15 t
asks are for partitions Vector(0))
26/09/24 22:42:08 INFO TaskSchedulerImpl: Adding task set 24.0 with 1 tasks resource profile 0
26/09/24 22:42:08 INFO TaskSetManager: Starting task 0.0 in stage 24.0 (TID 17) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:42:08 INFO Executor: Running task 0.0 in stage 24.0 (TID 17)
26/09/24 22:42:08 INFO ShuffleBlockFetcherIterator: Getting 1 (72.0 B) non-empty blocks including 1 (72.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) rem
ote blocks
26/09/24 22:42:08 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:42:08 INFO CodeGenerator: Code generated in 8.986675 ms
26/09/24 22:42:08 INFO Executor: Finished task 0.0 in stage 24.0 (TID 17), 3995 bytes result sent to driver
26/09/24 22:42:08 INFO TaskSetManager: Finished task 0.0 in stage 24.0 (TID 17) in 18 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:42:08 INFO TaskSchedulerImpl: Removed TaskSet 24.0, whose tasks have all completed, from pool
26/09/24 22:42:08 INFO DAGScheduler: ResultStage 24 (collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78) finished in 0.024 s
26/09/24 22:42:08 INFO DAGScheduler: Job 17 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:42:08 INFO TaskSchedulerImpl: Killing all running tasks in stage 24: Stage finished
26/09/24 22:42:08 INFO DAGScheduler: Job 17 Finished: collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:78, took 0.027013 s
Total Transactions: 1000
Total Quantity Sold: 25355
Total Calculated Revenue: 70329940.70999995
Average Sales Amount: 5019.265230000003
26/09/24 22:42:08 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:42:08 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:42:08 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:42:08 INFO MemoryStore: MemoryStore cleared
26/09/24 22:42:08 INFO BlockManager: BlockManager stopped
26/09/24 22:42:08 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:42:08 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:42:08 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:42:08 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:42:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-04aa601a-e0d5-4e90-8f02-affda2bddb71/pyspark-99bf635c-606a-4d3d-9a3b-7aa002f19c19
26/09/24 22:42:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-f74e7510-e213-43d3-af43-b421dc6ec05a
26/09/24 22:42:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-04aa601a-e0d5-4e90-8f02-affda2bddb71

(tushar@Tushar-Laptop): ~/Desktop/practical6_BDA

```

```
Session Actions Edit View Help
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_31_piece0 stored as bytes in memory (estimated size 22.1 KiB, free 433.3 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_31_piece0 in memory on 10.255.255.254:45405 (size: 22.1 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 31 from broadcast at DAGScheduler.scala:1611
26/09/24 22:43:23 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 27 (MapPartitionsRDD[77] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partition s Vector(0))
26/09/24 22:43:23 INFO TaskSchedulerImpl: Adding task set 27.0 with 1 tasks resource profile 0
26/09/24 22:43:23 INFO TaskSetManager: Starting task 0.0 in stage 27.0 (TID 19) in 13 ms on 10.255.255.254 (executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:43:23 INFO Executor: Running task 0.0 in stage 27.0 (TID 19)
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Getting 1 (336.0 B) non-empty blocks including 1 (336.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:43:23 INFO Executor: Finished task 0.0 in stage 27.0 (TID 19). 5175 bytes result sent to driver
26/09/24 22:43:23 INFO TaskSetManager: Finished task 0.0 in stage 27.0 (TID 19) in 13 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:43:23 INFO TaskSchedulerImpl: Removed TaskSet 27.0, whose tasks have all completed, from pool
26/09/24 22:43:23 INFO DAGScheduler: ResultStage 27 (showString at NativeMethodAccessorImpl.java:0) finished in 0.019 s
26/09/24 22:43:23 INFO DAGScheduler: Job 19 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:43:23 INFO TaskSchedulerImpl: Killing all running tasks in stage 27: Stage finished
26/09/24 22:43:23 INFO DAGScheduler: Job 19 finished: showString at NativeMethodAccessorImpl.java:0, took 0.022726 s

+-----+-----+
|Product_Category|Total_Revenue|
+-----+-----+
|Clothing|1.9286733929999996E7|
+-----+-----+
only showing top 1 row

Highest Revenue Region:
26/09/24 22:43:23 INFO FileSourceStrategy: Pushed Filters:
26/09/24 22:43:23 INFO FileSourceStrategy: Post-Scan Filters:
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_32 stored as values in memory (estimated size 199.5 KiB, free 433.1 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Block broadcast_32_piece0 stored as bytes in memory (estimated size 34.3 KiB, free 433.0 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_32_piece0 in memory on 10.255.255.254:45405 (size: 34.3 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 32 from showString at NativeMethodAccessorImpl.java:0
26/09/24 22:43:23 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/24 22:43:23 INFO DAGScheduler: Registering RDD 81 (showString at NativeMethodAccessorImpl.java:0) as input to shuffle 8
26/09/24 22:43:23 INFO DAGScheduler: Got map stage job 20 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:43:23 INFO DAGScheduler: Final stage: ShuffleMapStage 28 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:43:23 INFO DAGScheduler: Parents of final stage: List()
26/09/24 22:43:23 INFO DAGScheduler: Missing parents: List()
26/09/24 22:43:23 INFO DAGScheduler: Submitting ShuffleMapStage 28 (MapPartitionsRDD[81] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_33 stored as values in memory (estimated size 44.4 KiB, free 433.0 MiB)
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_33_piece0 stored as bytes in memory (estimated size 20.5 KiB, free 433.0 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_33_piece0 in memory on 10.255.255.254:45405 (size: 20.5 KiB, free: 434.1 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 33 from broadcast at DAGScheduler.scala:1611
26/09/24 22:43:23 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 28 (MapPartitionsRDD[81] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
```

```
Session Actions Edit View Help
26/09/24 22:43:23 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but current version of codegen fast hashmap does not support this aggregate.
26/09/24 22:43:23 INFO BlockManagerInfo: Removed broadcast_24_piece0 on 10.255.255.254:45405 in memory (size: 9.8 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Removed broadcast_29_piece0 on 10.255.255.254:45405 in memory (size: 34.3 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Starting job: showString at NativeMethodAccessorImpl.java:0
26/09/24 22:43:23 INFO DAGScheduler: Got job 21 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/24 22:43:23 INFO DAGScheduler: Final stage: ResultStage 30 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:43:23 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 29)
26/09/24 22:43:23 INFO DAGScheduler: Missing parents: List()
26/09/24 22:43:23 INFO DAGScheduler: Submitting ResultStage 30 (MapPartitionsRDD[85] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_34 stored as values in memory (estimated size 47.5 KiB, free 433.4 MiB)
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_34_piece0 stored as bytes in memory (estimated size 22.1 KiB, free 433.4 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_34_piece0 in memory on 10.255.255.254:45405 (size: 22.1 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 34 from broadcast at DAGScheduler.scala:1611
26/09/24 22:43:23 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 30 (MapPartitionsRDD[85] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partition s Vector(0))
26/09/24 22:43:23 INFO TaskSchedulerImpl: Adding task set 30.0 with 1 tasks resource profile 0
26/09/24 22:43:23 INFO TaskSetManager: Starting task 0.0 in stage 30.0 (TID 21) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:43:23 INFO Executor: Running task 0.0 in stage 30.0 (TID 21)
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Getting 1 (320.0 B) non-empty blocks including 1 (320.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:43:23 INFO Executor: Finished task 0.0 in stage 30.0 (TID 21). 5167 bytes result sent to driver
26/09/24 22:43:23 INFO TaskSetManager: Finished task 0.0 in stage 30.0 (TID 21) in 13 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:43:23 INFO TaskSchedulerImpl: Removed TaskSet 30.0, whose tasks have all completed, from pool
26/09/24 22:43:23 INFO DAGScheduler: ResultStage 30 (showString at NativeMethodAccessorImpl.java:0) finished in 0.020 s
26/09/24 22:43:23 INFO DAGScheduler: Job 21 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:43:23 INFO TaskSchedulerImpl: Killing all running tasks in stage 30: Stage finished
26/09/24 22:43:23 INFO DAGScheduler: Job 21 finished: showString at NativeMethodAccessorImpl.java:0, took 0.023562 s

+-----+-----+
|Region|Total_Revenue|
+-----+-----+
|North|1.8208403850000013E7|
+-----+-----+
only showing top 1 row

Top-Selling Product:
26/09/24 22:43:23 INFO FileSourceStrategy: Pushed Filters:
26/09/24 22:43:23 INFO FileSourceStrategy: Post-Scan Filters:
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_35 stored as values in memory (estimated size 199.5 KiB, free 433.2 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Block broadcast_35_piece0 stored as bytes in memory (estimated size 34.3 KiB, free 433.1 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_35_piece0 in memory on 10.255.255.254:45405 (size: 34.3 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 35 from showString at NativeMethodAccessorImpl.java:0
26/09/24 22:43:23 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/24 22:43:23 INFO DAGScheduler: Registering RDD 89 (showString at NativeMethodAccessorImpl.java:0) as input to shuffle 9
26/09/24 22:43:23 INFO DAGScheduler: Got map stage job 22 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
```

```
Session Actions Edit View Help
26/09/24 22:43:23 INFO DAGScheduler: Final stage: ResultStage 33 (showString at NativeMethodAccessorImpl.java:0)
26/09/24 22:43:23 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 32)
26/09/24 22:43:23 INFO DAGScheduler: Missing parents: List()
26/09/24 22:43:23 INFO DAGScheduler: Submitting ResultStage 33 (MapPartitionsRDD[93] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_37 stored as values in memory (estimated size 46.1 KiB, free 433.0 MiB)
26/09/24 22:43:23 INFO MemoryStore: Block broadcast_37_piece0 stored as bytes in memory (estimated size 21.6 KiB, free 433.0 MiB)
26/09/24 22:43:23 INFO BlockManagerInfo: Added broadcast_37_piece0 in memory on 10.255.255.254:45405 (size: 21.6 KiB, free: 434.2 MiB)
26/09/24 22:43:23 INFO SparkContext: Created broadcast 37 from broadcast at DAGScheduler.scala:1611
26/09/24 22:43:23 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 33 (MapPartitionsRDD[93] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partition s Vector(0))
26/09/24 22:43:23 INFO TaskSchedulerImpl: Adding task set 33.0 with 1 tasks resource profile 0
26/09/24 22:43:23 INFO TaskSetManager: Starting task 0.0 in stage 33.0 (TID 23) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:43:23 INFO Executor: Running task 0.0 in stage 33.0 (TID 23)
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Getting 1 (5.3 KiB) non-empty blocks including 1 (5.3 KiB) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/24 22:43:23 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:43:23 INFO Executor: Finished task 0.0 in stage 33.0 (TID 23). 5151 bytes result sent to driver
26/09/24 22:43:23 INFO TaskSetManager: Finished task 0.0 in stage 33.0 (TID 23) in 14 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:43:23 INFO TaskSchedulerImpl: Removed TaskSet 33.0, whose tasks have all completed, from pool
26/09/24 22:43:23 INFO DAGScheduler: ResultStage 33 (showString at NativeMethodAccessorImpl.java:0) finished in 0.019 s
26/09/24 22:43:23 INFO DAGScheduler: Job 23 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:43:23 INFO TaskSchedulerImpl: Killing all running tasks in stage 33: Stage finished
26/09/24 22:43:23 INFO DAGScheduler: Job 23 finished: showString at NativeMethodAccessorImpl.java:0, took 0.023127 s

+-----+-----+
|Product_ID|Total_Quantity_Sold|
+-----+-----+
|1090|590|
+-----+-----+
only showing top 1 row

26/09/24 22:43:23 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:43:23 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:43:24 INFO OutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:43:24 INFO MemoryStore: MemoryStore cleared
26/09/24 22:43:24 INFO BlockManager: BlockManager stopped
26/09/24 22:43:24 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:43:24 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:43:24 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:43:24 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:43:24 INFO ShutdownHookManager: Deleting directory /tmp/spark-8aaf381f-0328-42f1-b87a-950d26f3687e
26/09/24 22:43:24 INFO ShutdownHookManager: Deleting directory /tmp/spark-8aaf381f-0328-42f1-b87a-950d26f3687e/pyspark-bb042d32-6484-4a2a-b8a3-9ae89a2b12d5
26/09/24 22:43:24 INFO ShutdownHookManager: Deleting directory /tmp/spark-4cce2856-ec04-4037-9045-51b982d1b635
```

Key Questions / Analysis

1. Why are Spark DataFrames preferred over traditional spreadsheets for large e-commerce datasets?

Apache Spark DataFrames are preferred because they can process large datasets efficiently using distributed computing. They support operations such as filtering, grouping, aggregation, and sorting on large volumes of data, whereas traditional spreadsheets are more suitable for smaller datasets.

2. What is the role of groupBy() in e-commerce sales analysis?

The groupBy() operation groups records based on a particular column and allows aggregate calculations to be performed on each group. In this practical, it was used to calculate category-wise revenue, product-wise quantity sold, and region-wise revenue.

3. What is the difference between transformations and actions in Spark?

Transformations create a new DataFrame or RDD from an existing one, while **actions** execute the computation and return results. Examples of transformations used in this practical include withColumn(), groupBy(), and orderBy(), while show() and count() are examples of actions.

4. How does Spark help in processing large-scale e-commerce data?

Spark uses distributed processing to divide computational work across available resources. This allows operations such as aggregation, sorting, and analysis to be performed efficiently even when the dataset becomes much larger.

Supplementary Analysis

Find the top 3 revenue-generating cities.

```

Session Actions Edit View Help
26/09/24 22:48:41 INFO DAGScheduler: Missing parents: List()
26/09/24 22:48:41 INFO DAGScheduler: Submitting ResultStage 36 (MapPartitionsRDD[101] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/24 22:48:41 INFO MemoryStore: Block broadcast_40 stored as values in memory (estimated size 47.5 KiB, free 432.7 MiB)
26/09/24 22:48:41 INFO MemoryStore: Block broadcast_40_piece0 stored as bytes in memory (estimated size 22.1 KiB, free 432.7 MiB)
26/09/24 22:48:41 INFO BlockManagerInfo: Added broadcast_40_piece0 in memory on 10.255.255.254:38837 (size: 22.1 KiB, free: 434.1 MiB)
26/09/24 22:48:41 INFO SparkContext: Created broadcast 40 from broadcast at DAGScheduler.scala:1611
26/09/24 22:48:41 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 36 (MapPartitionsRDD[101] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for parti
ns Vector(0))
26/09/24 22:48:41 INFO TaskSchedulerImpl: Adding task set 36.0 with 1 tasks resource profile 0
26/09/24 22:48:41 INFO TaskSetManager: Starting task 0.0 in stage 36.0 (TID 25) (10.255.255.254, executor driver, partition 0, NODE_LOCAL, 8999 bytes)
26/09/24 22:48:41 INFO Executor: Running task 0.0 in stage 36.0 (TID 25)
26/09/24 22:48:41 INFO ShuffleBlockFetcherIterator: Getting 1 (320.0 B) non-empty blocks including 1 (320.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B)
remote blocks
26/09/24 22:48:41 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/24 22:48:41 INFO Executor: Finished task 0.0 in stage 36.0 (TID 25). 5265 bytes result sent to driver
26/09/24 22:48:41 INFO TaskSetManager: Finished task 0.0 in stage 36.0 (TID 25) in 12 ms on 10.255.255.254 (executor driver) (1/1)
26/09/24 22:48:41 INFO TaskSchedulerImpl: Removed TaskSet 36.0, whose tasks have all completed, from pool
26/09/24 22:48:41 INFO DAGScheduler: ResultStage 36 (showString at NativeMethodAccessorImpl.java:0) finished in 0.019 s
26/09/24 22:48:41 INFO DAGScheduler: Job 25 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:48:41 INFO TaskSchedulerImpl: Killing all running tasks in stage 36: Stage finished
26/09/24 22:48:41 INFO DAGScheduler: Job 25 finished: showString at NativeMethodAccessorImpl.java:0, took 0.022754 s
+-----+
|Region|      Total_Revenue|
+-----+
|North|1.8208403850000013E7|
|East|1.807740105000001E7|
|West|1.776156864999999E7|
+-----+
only showing top 3 rows

26/09/24 22:48:41 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:48:41 INFO SparkUI: Stopped Spark web UI at https://10.255.255.254:4040
26/09/24 22:48:41 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:48:41 INFO MemoryStore: MemoryStore cleared
26/09/24 22:48:41 INFO BlockManager: BlockManager stopped
26/09/24 22:48:41 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:48:41 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:48:41 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:48:42 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:48:42 INFO ShutdownHookManager: Deleting directory /tmp/spark-d18e7754-2ae9-478b-91cd-deecf15db6ef
26/09/24 22:48:42 INFO ShutdownHookManager: Deleting directory /tmp/spark-d18e7754-2ae9-478b-91cd-deecf15db6ef/pyspark-91557dfa-dbd1-4d22-b1c3-4c091be9a48e
26/09/24 22:48:42 INFO ShutdownHookManager: Deleting directory /tmp/spark-81574e7f-b56a-452c-ad77-f1f0498f4627

(tushar@Tushar-Laptop): [~/Desktop/practical6_BDA]
```

Calculate average revenue per order.

```
26/09/24 22:57:12 INFO TaskSchedulerImpl: Killing all running tasks in stage 37: Stage finished
26/09/24 22:57:12 INFO DAGScheduler: Job 27 finished: collect at /home/tushar/Desktop/practical6_BDA/sales_analysis.py:104, took 0.019332 s
Average Revenue per Transaction: 70329.94070999995
Least-Performing Product Category:
26/09/24 22:57:12 INFO FileSourceStrategy: Pushed Filters:
26/09/24 22:57:12 INFO FileSourceStrategy: Post-Scan Filters:
26/09/24 22:57:12 INFO MemoryStore: Block broadcast_44 stored as values in memory (estimated size 199.5 KiB, free 432.9 MiB)
26/09/24 22:57:12 INFO MemoryStore: Block broadcast_44 stored as values in memory (estimated size 199.5 KiB, free 432.9 MiB)
```

Least-Performing Product Category

```
26/09/24 22:53:27 INFO DAGScheduler: Job 29 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/24 22:53:27 INFO TaskSchedulerImpl: Killing all running tasks in stage 42: Stage finished
26/09/24 22:53:27 INFO DAGScheduler: Job 29 finished: showString at NativeMethodAccessorImpl.java:0, took 0.035225 s

+-----+-----+
|Product_Category|Total_Revenue|
+-----+-----+
|Food|1.5141885649999999E7|
+-----+-----+

only showing top 1 row

26/09/24 22:53:27 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:53:27 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:53:27 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:53:27 INFO MemoryStore: MemoryStore cleared
26/09/24 22:53:27 INFO BlockManager: BlockManager stopped
26/09/24 22:53:27 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:53:27 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:53:27 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:53:28 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:53:28 INFO ShutdownHookManager: Deleting directory /tmp/spark-05cf0a91-7e2b-4a04-9cd1-3616bfc23fde
26/09/24 22:53:28 INFO ShutdownHookManager: Deleting directory /tmp/spark-05cf0a91-7e2b-4a04-9cd1-3616bfc23fde/pyspark-6b5fe054-b116-41e6-b589-4117a3a95f6d
26/09/24 22:53:28 INFO ShutdownHookManager: Deleting directory /tmp/spark-c397ceaf-8e1f-43c7-8a78-6a200bb0f4ef

(tushar@Tushar-Laptop) ~/Desktop/practical6_BDA
$
```

Monthly Revenue Summary

```
26/09/24 22:55:03 INFO CodeGenerator: Code generated in 4.271863 ms

+-----+-----+
|Month|Total_Revenue|
+-----+-----+
|1|7488100.5099999998|
|2|5627477.4499999999|
|3|6008259.4100000004|
|4|5468497.6200000002|
|5|5800626.5099999998|
|6|6142561.23|
|7|4751432.4200000001|
|8|6716190.60999999985|
|9|4557015.58|
|10|6560606.3899999998|
|11|4958930.1899999997|
|12|6250242.79|
+-----+-----+

26/09/24 22:55:03 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:55:03 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:55:03 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:55:03 INFO MemoryStore: MemoryStore cleared
26/09/24 22:55:03 INFO BlockManager: BlockManager stopped
26/09/24 22:55:03 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:55:03 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:55:03 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:55:03 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:55:03 INFO ShutdownHookManager: Deleting directory /tmp/spark-89a29dc5-6ca2-49ef-b551-d510c249ade
26/09/24 22:55:03 INFO ShutdownHookManager: Deleting directory /tmp/spark-89a29dc5-6ca2-49ef-b551-d510c249ade/pyspark-24012334-c5ee-42c1-afb0-ba01ef095af6
26/09/24 22:55:03 INFO ShutdownHookManager: Deleting directory /tmp/spark-89a29dc5-6ca2-49ef-b551-d510c249ade

(tushar@Tushar-Laptop) ~/Desktop/practical6_BDA
$
```

Add a discount column and recompute revenue

```
26/09/24 22:57:12 INFO CodeGenerator: Code generated in 4.548646 ms

+-----+-----+-----+-----+
|Product_ID|Calculated_Revenue|Discount|Discounted_Revenue|
+-----+-----+-----+-----+
|1052|4809.960000000001|0.09|4377.063600000001|
|1093|71560.48|0.11|63688.8272|
|1015|11142.0|0.2|8913.6|
|1072|174242.25|0.02|170757.405|
|1061|9005.23|0.08|8284.811599999999|
|1021|35408.32|0.21|27972.5728|
|1083|76098.61|0.14|65444.8046|
|1087|182373.9|0.12|160489.032|
|1075|32863.499999999999|0.05|31220.324999999999|
|1075|48282.12|0.13|42005.4444|
+-----+-----+-----+-----+

only showing top 10 rows

26/09/24 22:57:12 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/09/24 22:57:12 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/24 22:57:12 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/24 22:57:12 INFO MemoryStore: MemoryStore cleared
26/09/24 22:57:12 INFO BlockManager: BlockManager stopped
26/09/24 22:57:12 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/24 22:57:12 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/24 22:57:12 INFO SparkContext: Successfully stopped SparkContext
26/09/24 22:57:13 INFO ShutdownHookManager: Shutdown hook called
26/09/24 22:57:13 INFO ShutdownHookManager: Deleting directory /tmp/spark-9b7bb028-84d7-489e-8cb2-7cecd30b9b82
26/09/24 22:57:13 INFO ShutdownHookManager: Deleting directory /tmp/spark-9b7bb028-84d7-489e-8cb2-7cecd30b9b82/pyspark-13777fbd-4f53-43eb-85e5-d5e428c705d5
26/09/24 22:57:13 INFO ShutdownHookManager: Deleting directory /tmp/spark-91e534f3-14d5-4d3e-aa36-7405d00bd4dc

(tushar@Tushar-Laptop) ~/Desktop/practical6_BDA
$
```

Tools and Technologies:

Programming Language: Python 3
Processing Framework: Apache Spark
Spark API: PySpark
Data Processing: Spark DataFrames
Input Data Format: CSV
Data Source: E-Commerce Sales Dataset
Data Analysis: Spark SQL / DataFrame Operations
Aggregation: groupBy(), agg(), sum(), avg()
Data Transformation: withColumn(), to_date(), month()
Sorting: orderBy()
Operating System: Ubuntu Linux / WSL2
Execution Environment: Local Spark (local[*])
Package Management: Python pip
Virtual Environment: Python venv
Development Environment: Ubuntu Terminal / Nano
Dataset File: ecommerce_sales.csv

Conclusion:

The e-commerce sales dataset was successfully processed using Apache Spark DataFrames. The dataset was loaded and its schema and data types were successfully validated. Revenue was calculated using Quantity Sold $\times$ Unit Price. Category-wise, product-wise, and region-wise sales analysis was performed using DataFrame operations. Additional analysis such as discount-adjusted revenue and monthly revenue was also performed. Thus, this practical demonstrated the use of Apache Spark for efficient e-commerce data analysis and business insights.

Github Repo:

<https://github.com/tushar-2606/BDA-SHP-2026>